

Find i_o in the circuit of Fig. 4.19 using source transformation.

Practice Problem 4.6

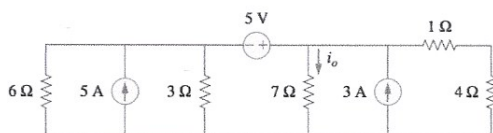


Figure 4.19
For Practice Prob. 4.6.

Answer: 1.78 A.

电源转换，简化电路再计算

$$① 6\Omega // 3\Omega = 2\Omega$$

$$② 5A // 2\Omega = 10V + 2\Omega \Rightarrow 15V + 2\Omega$$

$$③ 3A // 5\Omega = 15V + 5\Omega$$

$$④ \text{利用叠加原理. 只考虑左边的 } 15V \text{ 时, } i_o = \frac{15}{2 + \frac{35}{12}} \times \frac{5}{12} = \frac{75}{59} A$$

$$\text{只考虑右边的 } 15V \text{ 时, } i_o = \frac{15}{5 + \frac{14}{9}} \times \frac{2}{9} = \frac{30}{59} A$$

$$\therefore i_o = \frac{105}{59} A = 1.78 A$$

74

Practice Problem 4.12

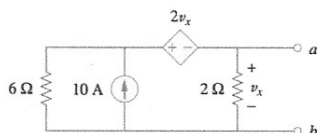
Find the Norton equivalent circuit of the circuit in Fig. 4.45 at terminals $a-b$.

Figure 4.45
For Practice Prob. 4.12.

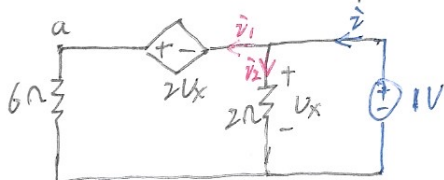
Answer: $R_N = 1\Omega$, $I_N = 10A$.

诺顿定理，含受控源：

①短路电流；②turn off独立源，外加测试电源，计算输入电阻

$$① \text{求短路电流. } v_x = 0, i_{sc} = 10A \Rightarrow I_N = 10A$$

$$② \text{求等效电阻. Turn off独立源, 施加激励, 求响应.}$$

施加 1V, 求 i

$$v_x = 1V \Rightarrow v_a = 3V$$

$$i_1 = \frac{3V}{6\Omega} = 0.5A$$

$$i_2 = \frac{1V}{2\Omega} = 0.5A$$

$$\left. \begin{array}{l} i_1 = 0.5A \\ i_2 = 0.5A \end{array} \right\} i = 1A$$

$$\therefore R_N = 1\Omega$$

75

Practice Problem 4.13

Determine the value of R_L that will draw the maximum power from the rest of the circuit in Fig. 4.52. Calculate the maximum power.

Answer: 4.222 Ω , 2.901 W.

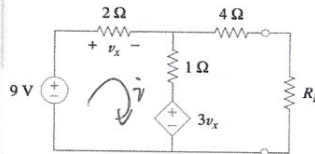
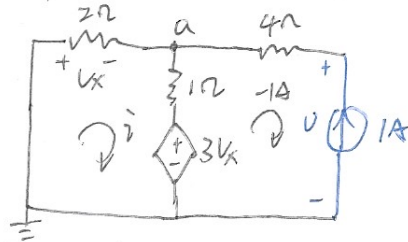


Figure 4.52
For Practice Prob. 4.13.

最大功率传输，先计算戴维南等效电路

① 计算开路电压 $-9 + v_x + i + 3v_x = 0$
 $v_x = 2i$
 $\Rightarrow v_x = 2V \Rightarrow v_{TH} = 7V$

② 计算等效电阻，Turn 掉独立源，施加 1A，求 v



$$\begin{cases} v_x + (i+1) \cdot 1 + 3v_x = 0 \\ -3v_x + (-1-i) \cdot 1 + (-4) + v = 0 \end{cases}$$

$$v_x = 2i$$

$$\Rightarrow v = \frac{38}{9} \Rightarrow R_{TH} = \frac{38}{9} \Omega$$

$$\therefore R_L = \frac{38}{9} \Omega = 4.222 \Omega$$

76

$$P_{max} = \frac{(\frac{v_{TH}}{2})^2}{R_L} = \frac{(3.5)^2}{\frac{38}{9}} = 2.901 W$$

Find the voltage across each of the capacitors in Fig. 6.20.

Answer: $v_1 = 75 V$, $v_2 = 75 V$, $v_3 = 25 V$, $v_4 = 50 V$.

电容串并联：串联电容的q相等

Practice Problem 6.7

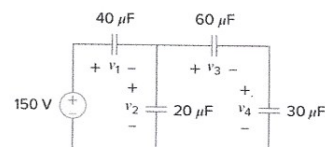


Figure 6.20
For Practice Prob. 6.7.

① $60\mu F$ 串联 $30\mu F = 20\mu F$

② $20\mu F \parallel 20\mu F = 40\mu F$

③ $40\mu F$ 与 $40\mu F$ 分压 $\Rightarrow v_1 = 75V$, $v_2 = 75V$

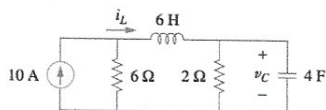
④ v_3 与 v_4 是 $60\mu F$ 与 $30\mu F$ 分压 $\Rightarrow v_3 = 75 \cdot \frac{1}{3} = 25V$

$$v_4 = 75 \cdot \frac{2}{3} = 50V$$

77

Practice Problem 6.10

Determine v_C , i_L , and the energy stored in the capacitor and inductor in the circuit of Fig. 6.28 under dc conditions.



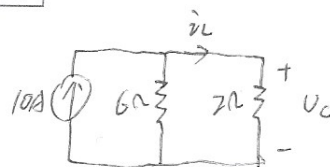
Answer: 15 V, 7.5 A, 450 J, 168.75 J.

电容直流开路；电感直流短路

Figure 6.28

For Practice Prob. 6.10.

① dc情况下，电容开路，电感短路 $\Rightarrow$



$$\therefore i_L = 10 \times \frac{3}{4} = 7.5 \text{ A}$$

$$v_C = i_L \times 2 \Omega = 15 \text{ V}$$

$$\textcircled{2} \quad w_C = \frac{1}{2} C v^2 = \frac{1}{2} \times 4 \times 15^2 = 450 \text{ J}$$

$$w_L = \frac{1}{2} L i^2 = \frac{1}{2} \times 6 \times 7.5^2 = 168.75 \text{ J}$$